

International Institute of Information Technology, Hyderabad
MA4.101-Real Analysis (Monsoon-2026)

Assignment 2

Total marks: 100

Due date: 6 October, 2026

Note 1

Each of the three number systems $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$ was built in the same way. One writes down a collection of formal expressions, and then *stipulates* when two of these expressions are to be regarded as denoting the same object.

System	Names	Stipulated equality
$\mathbb{Z}$	$a - b, \ a, b \in \mathbb{N}$	$a - b = c - d$ iff $a + d = c + b$
$\mathbb{Q}$	$a // b, \ a, b \in \mathbb{Z}, \ b \neq 0$	$a // b = c // d$ iff $ad = cb$
$\mathbb{R}$	$\text{LIM}_{n \rightarrow \infty} a_n, \ (a_n) \text{ Cauchy in } \mathbb{Q}$	$\text{LIM } a_n = \text{LIM } b_n$ iff $(a_n) \sim (b_n)$

Here $\sim$ is equivalence of sequences. The essential point is that **a name is not the object it names**, and one object has many names.

Definition [Well-defined rule]. A *rule from X to Y* is an assignment R which to each *name* ν of an element of X associates an expression $R(\nu)$. A rule R from X to Y is *well-defined* if both of the following hold.

- (W1) For every name ν of an element of X , the expression $R(\nu)$ is a legitimate name of an element of Y .
- (W2) For all names ν, μ of elements of X , $\nu = \mu \implies R(\nu) = R(\mu)$. This is the substitution property for the rule.

Proposition. Let R be a rule from X to Y . Then R is well-defined if and only if there is a function $f : X \rightarrow Y$ such that $f(x) = R(\nu)$ for every $x \in X$ and every name ν of x and in that case f is unique.

Proof. Suppose R is well-defined. Let $x \in X$. Every element of X has at least one name; choose one, say ν , and set $f(x) = R(\nu)$, which is an element of Y by (W1). This assigns exactly one value to x : if μ is any other name of x then $\nu = \mu$, so $R(\nu) = R(\mu)$ by (W2), and these are the same element of Y . So f is a function with the stated property.

Conversely, suppose such an f exists. Then for any name ν of x we have $R(\nu) = f(x) \in Y$, giving (W1); and if $\nu = \mu$ are names of the same x , then $R(\nu) = f(x) = R(\mu)$, giving (W2).

Uniqueness is immediate: two such functions agree at every x , since each equals $R(\nu)$ for any chosen name ν of x . $\square$

Note 2

Lemma P (componentwise membership). *Let p be an ordered pair, say $p = (x, y)$. Then for all sets X and Y ,*

$$p \in X \times Y \iff (x \in X \text{ and } y \in Y).$$

Moreover the components x and y are determined by p .

Proof. By definition, $p \in X \times Y$ holds if and only if $p = (x', y')$ for some $x' \in X$ and $y' \in Y$. If $x \in X$ and $y \in Y$ we may take $x' := x$ and $y' := y$. Conversely, if $p = (x', y')$ with $x' \in X$, $y' \in Y$, then $(x, y) = (x', y')$, so $x = x'$ and $y = y'$; hence $x \in X$ and $y \in Y$. The same equivalence,

$$(x, y) = (x', y') \iff (x = x' \text{ and } y = y'),$$

gives that the components are determined by p .

Question (1) [12 Marks]. Let I and J be sets, let A_α be a set for each $\alpha \in I$, and let B_β be a set for each $\beta \in J$.

(a) [3 Marks] Prove that

$$\left(\bigcup_{\alpha \in I} A_\alpha \right) \times \left(\bigcup_{\beta \in J} B_\beta \right) = \bigcup_{(\alpha, \beta) \in I \times J} (A_\alpha \times B_\beta).$$

(b) [3 Marks] Deduce that for all sets A, B, C, D ,

$$(A \cup C) \times (B \cup D) = (A \times B) \cup (A \times D) \cup (C \times B) \cup (C \times D).$$

(c) [3 Marks] Using (b), prove that

$$(A \times B) \cup (C \times D) = (A \cup C) \times (B \cup D)$$

holds *if and only if*

$$(A \subseteq C \text{ or } D \subseteq B) \quad \text{and} \quad (C \subseteq A \text{ or } B \subseteq D).$$

(d) [3 Marks] Prove that $(A \times B) \setminus (C \times D) = (A \setminus C) \times (B \setminus D)$ holds if and only if

$$(A \cap C = \emptyset \text{ or } B \subseteq D) \quad \text{and} \quad (A \subseteq C \text{ or } B \cap D = \emptyset).$$

About this question. Note that $(A \times B) \cup (C \times D)$ and $(A \cup C) \times (B \cup D)$ are in general different sets. A counterexample tells you *that* the identity fails. This question tells you *why* it fails, and *exactly when* it holds.

Question (2) [12 Marks]. Let A and B be sets. Call a subset $R \subseteq A \times B$ a *rectangle* if $R = S \times T$ for some $S \subseteq A$ and $T \subseteq B$.

(a) [6 Marks] Prove that R is a rectangle if and only if it has the following closure property:

$$\text{whenever } (x, y) \in R \text{ and } (x', y') \in R, \text{ also } (x, y') \in R.$$

(b) [6 Marks] For $x \in A$ define the *slice* $R_x := \{y \in B : (x, y) \in R\}$. Prove that R_x is a set, that

$$R = \bigcup_{x \in A} (\{x\} \times R_x),$$

and that R is a rectangle if and only if any two non-empty slices of R are equal.

About this question. A subset of $A \times B$ is the same thing as a *relation* from A to B . The relation “ $x < y$ ” on $\mathbb{N}$, the graph of a function from A to B , the set of pairs of students who sit next to each other: all of these are subsets of a Cartesian product. Very few of them are themselves Cartesian products $S \times T$. This question tells you

exactly which ones are.

Draw a picture, with A along the horizontal axis and B along the vertical one. The property in (a) then says: whenever two points of R are opposite corners of a rectangle, the other two corners are in R as well. What makes (a) useful is that it is an *internal* test: you can decide whether R is a product by looking at R alone, without being told what S and T are. That is also what makes the “if” direction harder than it first looks. You are not handed S and T ; you have to build them out of R , and only then can you check that $R = S \times T$.

Part (b) cuts R into vertical slices. The displayed identity says that every subset of $A \times B$ is the union of its slices, so a subset of $A \times B$ carries exactly the same information as a choice of a subset $R_x \subseteq B$ for each $x \in A$. Rectangles are the subsets for which all these choices agree, apart from the empty ones. Ask yourself why the words “non-empty” cannot be left out.

Question (3) [10 Marks]. Let $f : X \rightarrow Y$ and $g : Z \rightarrow W$ be functions. Define

$$f \times g : X \times Z \rightarrow Y \times W, \quad (f \times g)(x, z) := (f(x), g(z)).$$

- (a) **[5 Marks]** Prove that this is a function, and that $(f \times g)(A \times B) = f(A) \times g(B)$ for all $A \subseteq X$ and $B \subseteq Z$.
- (b) **[5 Marks]** Prove that $(f \times g)^{-1}(U \times V) = f^{-1}(U) \times g^{-1}(V)$ for all $U \subseteq Y$ and $V \subseteq W$.

About this question. So far $\times$ has been an operation on sets. Here it becomes an operation on *functions*: two functions acting separately on two inputs combine into a single function acting on pairs, such as $(x, z) \mapsto (x^2, z + 1)$. This is the form in which products appear later, for example when one studies functions of two variables.

The first claim, that $f \times g$ is a function at all, is where Note 2 is used. To compute $(f \times g)(p)$ you must first extract the two components of the pair p , and this extraction must give only one answer.

Question (4) [Exercise 3.6.5 of Tao] [6 Marks]. Let A and B be sets. Show that $A \times B$ and $B \times A$ have equal cardinality by constructing an explicit bijection between the two sets.

About this question. The sets $A \times B$ and $B \times A$ are usually *different*: $\{0\} \times \{1\}$ contains the pair $(0, 1)$, while $\{1\} \times \{0\}$ contains the pair $(1, 0)$, and these two pairs are not equal. Yet this question shows that the two sets always have the same number of elements.

Question (5) [Exercise 3.6.6 of Tao] [10+2 Marks]. Let A, B, C be sets. Show that the sets $(A^B)^C$ and $A^{B \times C}$ have equal cardinality by constructing an explicit bijection between the two sets. Conclude that $(a^b)^c = a^{bc}$ for any natural numbers a, b, c .

About this question. An element of $A^{B \times C}$ is a function of two variables: it takes a pair (b, c) and returns an element of A . An element of $(A^B)^C$ is a function of one variable whose *values are functions*: it takes $c \in C$ and returns a function from B to A . The question says that these are two ways of recording the same information. For example, the rule $(b, c) \mapsto b + c$ on $\mathbb{N} \times \mathbb{N}$ can equally be read as “given c , return the function $b \mapsto b + c$ ”. Passing between the two points of view is called *currying*, and it is used constantly in mathematics and in programming.

Question (6) [Exercise 3.6.9 of Tao] [10 Marks]. Let A and B be finite sets. Show that $A \cup B$ and $A \cap B$ are also finite sets, and that

$$|A| + |B| = |A \cup B| + |A \cap B|.$$

About this question. This is the two-set case of *inclusion–exclusion*. When you add $|A|$ and $|B|$, every element of $A \cap B$ is counted twice. The identity says that adding $|A \cap B|$ to $|A \cup B|$ produces exactly the same overcount.

Question (7) [Exercise 3.6.10 of Tao] [10 Marks]. Let $A_1, \dots, A_n$ be finite sets such that

$$\left| \bigcup_{i \in \{1, \dots, n\}} A_i \right| > n.$$

Show that there exists $i \in \{1, \dots, n\}$ such that $|A_i| \geq 2$. This is known as the pigeonhole principle.

About this question. In everyday language: if more than n objects are put into n boxes, then some box receives at least two. Here A_i is the set of objects in the i th box.

The statement looks too obvious to need a proof, but “obvious” here really means “we know how to count”, and in this course counting has been *defined* through bijections with sets $\{1, \dots, m\}$. So the statement has to be proved from that definition. The natural thing to prove, by induction on n , is the contrapositive: if every box contains at most one object, then altogether there are at most n objects. Notice that the sets A_i are not assumed to be disjoint, so one object may sit in several boxes.

The principle is remarkable for what it does not tell you: it proves that some box is crowded without saying which one. Despite its simplicity it has strong consequences. For example, among any $n + 1$ natural numbers, two leave the same remainder on division by n .

Question (8) [Exercise 4.1.8 of Tao] [7 Marks]. Show that the principle of induction (Peano Axiom 5) does not apply directly to the integers. More precisely, give an example of a property $P(n)$ pertaining to an integer n such that $P(0)$ is true, and that $P(n)$ implies $P(n++)$ for all integers n , but that $P(n)$ is not true for all integers n .

About this question. Induction works on $\mathbb{N}$ because every natural number is reached from 0 by applying $++$ finitely many times; Peano Axiom 5 is exactly the statement that $\mathbb{N}$ contains nothing else. The hypothesis “ $P(n)$ implies $P(n++)$ ” only pushes truth *forwards*. In $\mathbb{Z}$ there are elements that can never be reached from 0 by moving forwards, and the hypothesis says nothing at all about them.

So this question tells you that induction is not a general law of logic, valid in any number system. It is a special property of $\mathbb{N}$, and in fact it is the property that separates $\mathbb{N}$ from larger systems such as $\mathbb{Z}$.

Question 9 [6 Marks]. Answer the following questions.

1. **[3 Marks]** Let integers $\mathbb{Z}$ are defined as collection of names $a \text{ --- } b$, where $a \text{ --- } b = c \text{ --- } d \Leftrightarrow a + d = b + c$ and $a, b, c, d \in \mathbb{N}$. Consider a rule from $\mathbb{Z}$ to $\mathbb{N}$ given by

$$\text{ignore}(a \text{ --- } b) := a,$$

In words: **ignore** assigns to a name its first coordinate. Show that **ignore** is not well-defined.

2. **[3 Marks]** Let rationals $\mathbb{Q}$ are defined as collection of names $a \text{ // } b$, where $a \text{ // } b = c \text{ // } d \Leftrightarrow ad = bc$ and $a, b, c, d \in \mathbb{Z}$ and $b \neq 0$ and $d \neq 0$. Consider a rule

from $\mathbb{Q} \times \mathbb{Q}$ to $\mathbb{Q}$ given by

$$\text{mediant}(a \parallel b, c \parallel d) := (a + c) \parallel (b + d).$$

Show that **mediant** is not a well defined operation.

About this question. Every integer and every rational number has infinitely many names; for instance $3 \parallel 4 = 6 \parallel 8 = (-3) \parallel (-4)$. A rule is written in terms of a name, and nothing guarantees that it gives the same answer for two names of the same number.

Note that **mediant** is a sensible and useful rule on *fractions*: it is how the Farey sequences are built, which list in increasing order the fractions between 0 and 1 whose denominators are at most a given number.

Question (10) [6 Marks]. Call a sequence (a_n) *rigid* if

there is $N \geq 1$ such that for every $\varepsilon > 0$, $|a_j - a_k| \leq \varepsilon$ for all $j, k \geq N$.

- (a) **[3 Marks]** Prove that (a_n) is rigid if and only if it is *eventually constant*, that is, there is $N \geq 1$ with $a_j = a_N$ for all $j \geq N$.
- (b) **[3 Marks]** Deduce that every rigid sequence is Cauchy, and show that the converse fails.

About this question. Put the definition of a rigid sequence next to definition of a Cauchy sequence. Cauchy says: *for every $\varepsilon > 0$ there is an N such that ...* Rigid says: *there is an N such that for every $\varepsilon > 0$...* Only the order of two quantifiers has changed, and this question shows how drastic that change is. In the Cauchy definition, N is allowed to depend on ε . In the rigid definition a single N must serve every tolerance at once.

In analysis the order of the quantifiers *is* the content of a definition. Writing a definition from memory with two quantifiers swapped produces a different notion, and here a useless one: if real numbers were built from rigid sequences instead of Cauchy sequences, nothing beyond $\mathbb{Q}$ would come out, not even a square root of 2.

Question (11) [9 Marks]. Let $(a_n)_{n=1}^\infty$ be a Cauchy sequence of rationals.

- (a) **[3 Marks]** Let φ be any map from $\{n \in \mathbb{N} : n \geq 1\}$ to itself such that $\varphi(n) \geq n$ for every n . Prove that $(a_{\varphi(n)})_{n=1}^\infty$ is Cauchy and is equivalent to (a_n) .

- (b) [**3 Marks**] Deduce that (a_{n+1}) , (a_{2n}) and (a_{n^2}) are Cauchy and all name the same real number as (a_n) .
- (c) [**3 Marks**] Show that the hypothesis $\varphi(n) \geq n$ cannot simply be dropped, by exhibiting a Cauchy (a_n) and a map φ for which $(a_{\varphi(n)})$ is Cauchy but not equivalent to (a_n) .

About this question. A real number is named by a Cauchy sequence, and this question shows that the name does not depend on the early terms of the sequence. You may discard the first term, keep only every second term, or keep only the terms at square positions, and you still name the same real number.